

ASSIGNMENT 1- REPORT

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1. Overview

In this assignment, I focused on building a strong foundation in Python for data science. The goal was to move beyond basic programming and understand how to handle data efficiently using libraries like NumPy and Pandas.

2. Key Learnings & Implementation

Python Basics

I started by revisiting core Python structures. I worked with **Lists** (for ordered data) and **Dictionaries** (for key-value pairs) to understand when to use which. I also implemented control flow using loops and wrote modular functions to keep the code clean.

NumPy (Numerical Python)

The main focus here was working with arrays, which are faster than standard Python lists.

- **Initialization:** I learned to create arrays manually and generate them using random numbers or zeros.
- **Slicing:** I practiced slicing 1D and 2D arrays (e.g., `arr[:3, 1:]`) to extract specific rows and columns.
- **Dimensions:** Used `.shape` to check array sizes, which is helpful for debugging matrix operations.

Pandas (Data Manipulation)

I used Pandas to handle structured data (like Excel sheets).

- **DataFrames:** Created datasets from scratch using lists and dictionaries.
- **Selection:** I learned the difference between `loc` (label-based) and `iloc` (position-based) indexing.
- **Filtering:** I applied conditions to find specific data points, such as retrieving a specific player's stats without manually searching through the table.

3. Conclusion

This project helped me get comfortable with the syntax and logic needed for data analysis. Understanding how to slice arrays and filter DataFrames is essential for the more advanced machine learning tasks I plan to do next.